

Did You Know?

The first database systems were developed in the 1960s. Today, databases are used in almost every application and website.



Introduction to Database

A database is an organized collection of data that can be stored, managed and accessed easily. Databases help organizations store large amounts of information efficiently.



What is DBMS?

DBMS (Database Management System) is software used to create, store, manage and retrieve data from databases.

Advantages of DBMS

- Reduces Data Redundancy
- Improves Data Security
- Provides Data Consistency
- Allows Data Sharing
- Easy Data Management



Database Architecture

Database architecture describes how users interact with databases. The main levels are:

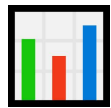
- External Level
- Conceptual Level
- Internal Level



Database Keys

Keys are used to identify records and create relationships between tables.

- **Primary Key:** Uniquely identifies each record.
- **Foreign Key:** Connects two tables.
- **Candidate Key:** Possible primary key.
- **Composite Key:** Combination of multiple attributes.



ER Diagram

Entity Relationship Diagram (ERD) is used to represent entities, attributes and relationships in a database.

- Entity
- Attribute
- Relationship
- Cardinality



SQL Basics

SQL (Structured Query Language) is used to communicate with databases.

Common SQL Commands

- SELECT - Retrieve Data
- INSERT - Add Data
- UPDATE - Modify Data
- DELETE - Remove Data
- CREATE - Create Database Objects



SQL Joins

Joins are used to combine data from multiple tables.

- INNER JOIN
- LEFT JOIN
- RIGHT JOIN
- FULL JOIN



Normalization

Normalization is the process of organizing data to reduce redundancy and improve database efficiency.

- 1NF - First Normal Form
- 2NF - Second Normal Form
- 3NF - Third Normal Form
- BCNF - Boyce Codd Normal Form



Database Security

- Authentication
- Authorization
- Encryption
- Backup and Recovery



Database Interview Questions

Q1: What is a Database?

A database is an organized collection of data.

Q2: What is DBMS?

DBMS is software used to manage databases.

Q3: What is Primary Key?

A primary key uniquely identifies each record in a table.

Q4: What is Normalization?

Normalization organizes data to reduce redundancy.

Final Quick Revision

Full Form: Database Management System

Language: SQL

Main Concepts: Tables, Keys, Joins, Normalization

Purpose: Store and Manage Data